NATHAN ZHANG

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RESEARCH EXPERIENCE

Research Assistant, UBC Sauder School of Business (Apr 2026 – Present)
Supervisors: J. Frank Li and Cheng Chen

- As sole RA on a study of age-biased technological change, link CPS, O*NET, and USPTO data to construct technological-exposure measures. Select fixed effects and fit weighted FE OLS and GMM models for wage and occupational sorting analysis.
- Harmonized 60+ historical O*NET releases. Build semantic measures with PatentsBERTa, use Qwen for robustness checks, and validate exposure through sampled manual reading of task and patent text.
- Implemented bootstrap inference, calibration, and robustness checks. Maintain the R/Python replication package and Bash/Slurm HPC workflows, and draft the paper's empirical section.

Research Assistant, UBC Sauder School of Business (May 2026 – Aug 2026)
Supervisor: Allen Hu

- Evaluated financial-news coverage and missingness through article-level matching of RavenPack metadata against a ProQuest corpus of over 3.2 million articles.
- Built yt-dlp workflows to support multimodal financial information extraction and analysis across media: clean panels from metadata, video/audio archiving and deduplication, and subtitle-availability checks.

Research Assistant, Vancouver School of Economics at UBC (Jan 2025 – May 2026)
Supervisors: Jonathan Graves and Laura Nelson

- Led CV, NLP, and NLI proposals for COMET and Praxis UBC, and implemented models, training, and explanatory text for a coauthored Kouroi classification notebook. The platforms have served 1,000+ cumulative student users.
- Co-developed a zero-shot stance-classification tutorial with human-label validation, in first revision at Programming Historian. Designed and taught a computer vision workshop.

PUBLICATIONS AND MANUSCRIPTS

Memory Response Paths and Path-Structured Context Compilation (2026)
Joint first author with Jerrold Huang. Submitted to AAAI and under review
Proposed response-path modeling of LLM memory and built benchmarks and model API experiments for statistical evaluation. In a separate 144-item evaluation, PSCC's LLM-judged quality is 6.94 percentage points higher than A-Mem's (95% CI: 2.64 to 11.32) while using 67.2% of its API tokens.

Sequential Tone Residualization and Abnormal Returns in Earnings Calls (2026)
Joint first author with Jerrold Huang. Accepted at ICAIF 2026, selected for oral presentation
Proposed the earnings-call application and dictionary/LLM validation, built a Capital IQ/IBES/CRSP event panel for 2018-2025 with historical S&P 500 membership, and co-designed stage-specific information sets. Time-holdout tests linked answer surprise positively to subsequent cumulative abnormal returns, with pre-event checks.

EDUCATION

The University of British Columbia (Sep 2023 – May 2028)
B.A., Combined Major in Economics and Statistics, Minor in Data Science, Vancouver, BC

- Cumulative average: 93.5/100

RESEARCH INTERESTS

Methods: Statistical learning and inference, causal inference and econometrics, panel and time-series analysis, text and image methods.
Applications: Technological change and labor, financial risk and ESG, computational social science.

SELECTED PROJECTS

Double Machine Learning for Payment-Friction Spending Effects (2025)
Course research project [Code] [Report]

- Contributed EDA, leakage checks, multi-model refitting, and econometric input on estimands to a collaborative DoubleML analysis of payment friction and spending across about 5,600 respondents and roughly 22,000 in-person person-days. The study also explored heterogeneous effects with DR Learner.

Canadian Inflation Forecasting and Macro Signal Evaluation (2025)
Forecasting project [Code]

- Built an R/Quarto VAR forecast for Canadian inflation with rolling out-of-sample tests against naive and random-walk baselines. Presented impulse responses in a Bank of Canada-style policy briefing.

TEACHING AND RESEARCH INFRASTRUCTURE

Image Analysis Workshop: Computer Vision for Social Science Research (Mar 2026)
Workshop lead and instructor [Workshop]
Designed and taught image embeddings, classification, and detection for social science research.

A Study of Richter's Kouroi Through Image Embedding (Jul 2025)
Coauthor with Yash Mali and Krishaant Pathman [Notebook]
Used ConvNeXt V2 embeddings, PCA, clustering, and classification to test whether image embeddings recover structure in Richter's kouroi categories.

LLM Prediction Module (2025)
Coauthor [Module]
Coauthored course material on next-word prediction, token probabilities, and economics applications.

PROFESSIONAL EXPERIENCE

ESG Research Analyst Co-op, Capital Markets & Credit Investments (Sep 2026 – Present)
British Columbia Investment Management Corporation (BCI), Vancouver, BC

- Investigate executive compensation and shareholder alignment for fundamental investment research. Independently construct variables from Capital IQ and ISS ECA and build Databricks pipelines.
- Automate voting rules, reconstruct ballot-time information, and reproduce research findings. Backtest against historical votes and subsequent risk outcomes to help the proxy team assess voting strategy.

Teaching Assistant (Sep 2024 – Aug 2025)
UBC Chinese Language Program, Vancouver, BC

- Supported Chinese language teaching, grading, volunteer training, and feedback coordination.

AWARDS AND SCHOLARSHIPS

Work Learn International Undergraduate Research Awards (WLIURA) (UBC), 2026
Economics Undergraduate Scholarship (Vancouver School of Economics, UBC), 2025
Faculty of Arts International Student Scholarship (UBC Faculty of Arts), 2024, 2025

SELECTED COURSEWORK

Statistical Inference (100), Machine Learning and Data Mining (93), Causal Methods (98)
Advanced Econometrics (98), Statistical Learning (97), Time Series and Forecasting (95)

TECHNICAL SKILLS AND LANGUAGES

Programming & data: Python, R, SQL, Bash, Julia, PySpark, DuckDB, Polars, BQL
Methods: econometrics, factor models, causal inference, panel/time series, machine learning, NLP, computer vision
Modeling libraries: NumPy, SciPy, scikit-learn, PyTorch, statsmodels, fixest, OpenCV, spaCy, NLTK
Platforms & tools: Databricks, Bloomberg, Claude Code, DevOps/Git, GitHub Actions, HPC/Slurm, Jupyter/Quarto
Languages: Mandarin Chinese: native / English: academic working proficiency / Japanese: JLPT N3